Nicole Huber, Phd

Research Scientist · Forensic Genetics & Mitochondrial DNA Phylogenetics

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PROFILE

Postdoctoral Research Scientist with expertise in forensic genetics, mitochondrial DNA phylogenetics, and bioinformatics. Track record of leading independent research projects, developing open-source computational frameworks, securing competitive funding, and publishing in peer-reviewed international journals.

WORK EXPERIENCE

10/2022 – present **Postdoctoral Researcher & Principal Investigator**

Institute of Legal Medicine, Medical University of Innsbruck

- Lead independent research project on human mitochondrial DNA phylogeny and forensic genetics applications.
- Secured competitive funding from FWF.
- Design and maintain bioinformatic pipelines for mtDNA sequence processing, quality control and phylogenetic analysis.
- Develop and publish open-source software (mitoLEAF) for mtDNA analysis.
- Curate and manage complex genomic datasets, ensuring quality and accessibility for research collaborations.
- Establish and maintain international research collaborations.

03/2021 – 03/2022 **Software Test Engineer**

MED-EL Medical Electronics, Innsbruck

- Software quality assurance and test automation for medical software.

12/2019 – 12/2020 **Educational leave**

- Educational leave for personal development and language training.

02/2015 – 12/2019 **Scientific Researcher / PhD Candidate**

Institute of Legal Medicine, Medical University of Innsbruck

- PhD research on mitochondrial DNA phylogenetics and forensic databases.
- Independent forensic genetics research on mtDNA analysis and database development.
- Applied bioinformatic pipelines and phylogenetic methods for QC and mixture resolution.
- Published peer-reviewed articles and contributed to mtDNA community databases.

11/2012 – 01/2015 **Software Test Engineer**

Biocrates Life Sciences AG, Innsbruck

- Software quality assurance and testing for metabolomic research software.

EDUCATION

11/2016 – 09/2019 **Ph.D. in Genetics & Genomics (with distinction)**

Medical University of Innsbruck

03/2013 – 09/2016 **M.Sc. in Forensic Science (with distinction)**

Staffordshire University, UK

10/2009 – 10/2012 **B.Sc. in Biomedical Informatics**

UMIT, Hall in Tirol

FUNDING & GRANTS

Project lead — **Austrian Science Fund (FWF) — Project ESP 222**
Research on mitochondrial DNA phylogenetics and forensic applications · DOI: 10.55776/ESP222

RESEARCH & TECHNICAL EXPERTISE

Research: Mitochondrial DNA phylogenetics and phylogeographic analysis · Forensic genetics (haplogrouping, mixture resolution, DNA databases) · Next-generation sequencing data analysis and QC · Reproducible research pipelines and workflow automation · Open science and scientific software development.

Programming: Python, R, Unix Shell, Java, C#

Bioinformatics: Sequence alignment, phylogenetic tree building, quality control

Databases & Infrastructure: MySQL, EMPOP, HPC environments, Git

PUBLICATIONS

Author on several peer-reviewed publications in forensic genetics and bioinformatics. Lead author on the mitoLEAF open-source framework (NAR Genomics & Bioinformatics, 2025). Full publication list provided separately.

SELECTED CONFERENCE CONTRIBUTIONS

Oral presentations

- Haploid Marker Meeting, Belgrade (05/2025); *"From mitoTree to mitoLEAF: advancing open science in mitochondrial DNA phylogenetics".*
- Haploid Marker Meeting, Bydgoszcz (05/2018); *"Search, Align and Haplogroup: improved forensic mtDNA analysis via EMPOP".*

Poster presentations

- MCEB, Granada (05/2025); *"mitoTree: a modern open-source framework for advancing mitochondrial DNA phylogenetics".*
- ISFG Congress, Santiago de Compostela (09/2024); *"mitoTree: advancing mtDNA research through updated phylogeny and haplogroup guidelines".*
- ISFG Congress, Prague (09/2019); *"Introducing a forensic version of Phylotree for improved haplogrouping".*

Full conference contributions and visits available upon request.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

10/2025 — **Introduction to Machine Learning with Python**
Swiss Institute of Bioinformatics (SIB), in person

05/2022 — **Python for Genomic Data Science**
Johns Hopkins University, online

09/2021 — **Whole Genome Sequencing of Bacterial Genomes: Tools and Applications**
Technical University of Denmark (DTU), online

LANGUAGES

German (native) · **English** (fluent) · **Spanish** (basic) · **Latin** (basic)